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Development of a New Two-Motor Plug-In Hybrid System

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ABSTRACT

A highly efficient two-motor plug-in hybrid system is developed to satisfy the global demands of CO_2 reduction. This system switches three operation modes, what is called "EV Drive", "Hybrid Drive" and "Engine Drive", to maximize fuel efficiency according to the driving condition of the vehicle. Practical plug-in EV (Electric Vehicle) capability is also realized by adding a high-power on-board charger and a high capacity Li-ion battery to the original system.

The outlines of the system components including a newly developed Atkinson cycle engine, a highly efficient electric coupled CVT (Continuously Variable Transmission) with built-in motor and generator, an integrated PCU (Power Control Unit) and an exclusive battery for plug-in HEV (Hybrid Electric Vehicle) are described in this paper.

In addition to the switching of three driving modes and the efficiency improvement of each device, cooperative control of the hybrid system is introduced.

By installing this system in 2014 Accord plug-in hybrid, superior fuel economy is achieved with values of 46 MPG in CS (Charge Sustaining) mode and 115 MPGe (Miles Per Gallon equivalent) in CD (Charge Depleting) mode. 13miles of AER (All Electric Range) is also achieved while maintaining sufficient driving performance as a mid-size sedan.

MPGe is a unit of fuel economy for the alternate fuel vehicles such as plug-in hybrid or electric vehicles. The ratings are calculated by converting used electric energy into equivalent amount of gasoline.

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INTRODUCTION

Compliance with global strengthening of emission regulations is an urgent issue for the automobile industry. The final solution for this requirement should be to putting zero-emission vehicles, such as BEV (Battery Electric Vehicle) or FCEV (Fuel Cell Electric Vehicle) into practical use. However, they could not be a suitable replacement of conventional cars because of their limited driving range and higher cost due to current technical limitations. Therefore, HEV (Hybrid Electric Vehicle) application has been expanding globally as one of practical solutions.

A new two-motor plug-in hybrid system which realizes both environmental friendliness of zero-emission vehicles and practicalities of HEVs is developed to satisfy this situation and main features are discussed in this paper.

OVERALL CONFIGURATION

The major components of the new two-motor plug-in hybrid system are placed in the vehicle as shown in Figure 1. An electric coupled CVT (e-CVT) is installed in the position of a conventional transmission and utilized in combination with an Atkinson cycle engine optimized for HEV application. The PCU (Power Control Unit) is positioned above the e-CVT and cooled by an independent water cooling circuit separated from the engine. Various functions including high-voltage inverters, VCU (Voltage Control Unit) and motor/generator ECU (Electric Control Unit) are integrated in the PCU. IPU (Intelligent Power Unit) is installed behind the rear seats and cooled by cooling fans located in the trunk room. A Li-ion battery specially designed for plug-in hybrid, a DC/DC converter, a high-power on-board charger and a battery ECU constitute the IPU as one package. Figure 2 shows a structure of the IPU.

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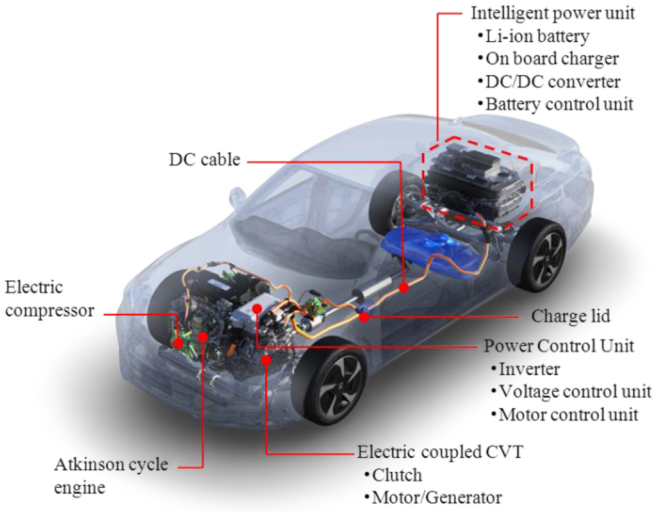


Figure 1. Overall system configuration

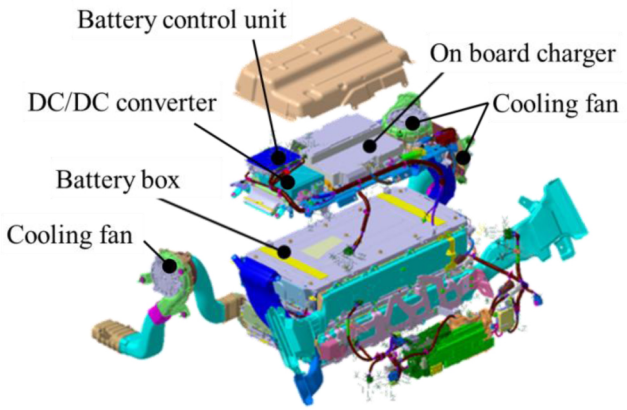


Figure 2. IPU (Intelligent Power Unit)

POWER TRAIN

Overview

An overview of the power train is shown in Figure 3 and operation modes of this system are explained in Table 1.

This system can be operated in three operation modes explained below.

In "EV Drive" mode, the electric motor physically connected to the driveshaft propels the vehicle by using electric energy stored in the high voltage battery. In "Hybrid Drive" mode, the electric motor is driven by electric energy generated by the generator using output from the engine. In "Engine Drive" mode, the engine-drive clutch located between the engine and wheels is engaged, and the vehicle is directly propelled by the engine output.

Fuel efficiency of all driving conditions is maximized by switching these three driving modes properly.

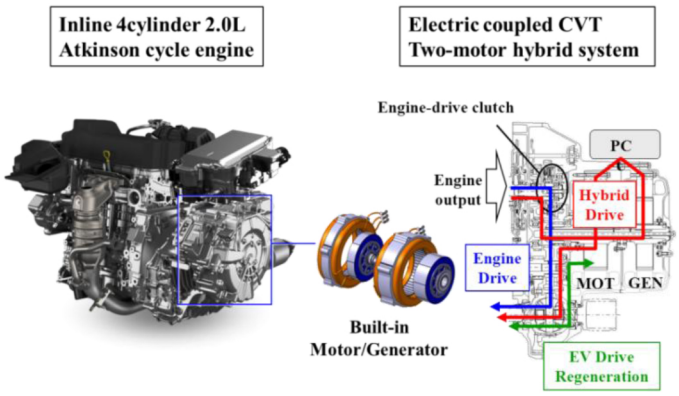
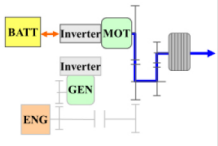
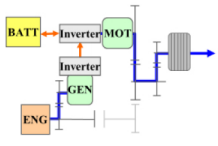
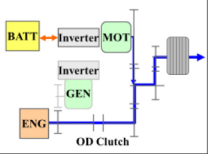


Figure 3. Hybrid power train

Table 1. Operation modes

Electrical Transfer
Mechanical Transfer

EV Drive	Hybrid Drive	Engine Drive
		
Use electric energy from battery to drive motor	Use electric energy from engine generator to drive motor, with assistance from battery	Use engine to directly drive wheels

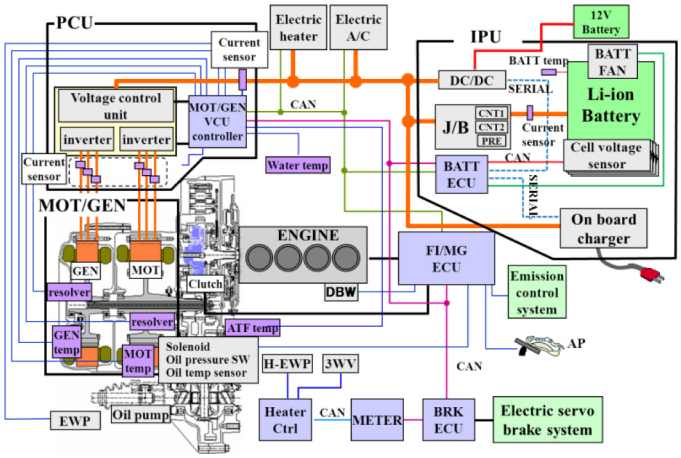


Figure 4. Block diagram of hybrid system

Figure 4 is a block diagram of this system. The powertrain system of this vehicle consists of an Atkinson cycle engine, an e-CVT with built-in motor and generator, a PCU and a high-voltage battery. ECUs, which control these power train components, realize cooperative control each other through the on-board communication network.

Moreover, an overall system configuration is designed for cooperative control with related systems such as an electric servo brake, an electric air conditioner and an electric heater

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to optimize system efficiency according to the situation of the vehicle.

The outlines of the powertrain components and related systems are described in the following sections.

Atkinson cycle engine

A 2.0 L DOHC (Double Over Head Camshaft) engine for this two-motor hybrid system is developed newly. Atkinson cycle and cooled EGR (Exhaust Gas Recirculation system) are adopted and these technologies combine with the VTEC (Variable valve Timing and Electric Control) system. This engine has two cam profiles, output cam and fuel economy (FE) cam, so maximum output 105 kW and 10% improvement of BSFC are attained.

VTEC system and operation area are shown in Figure 5 and 6. Engine power performance and enhancement of fuel consumption are shown in Figure 7 and 8.

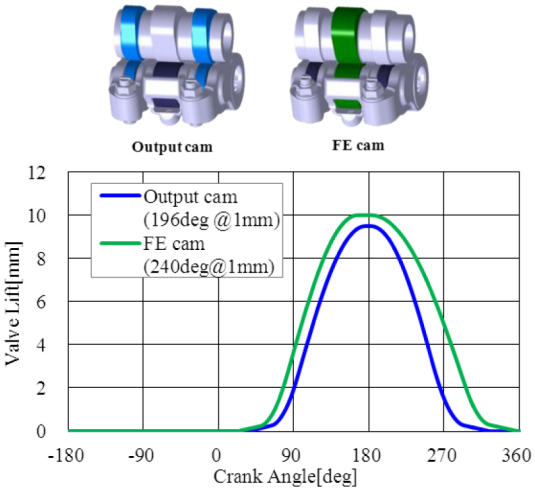


Figure 5. VTEC system

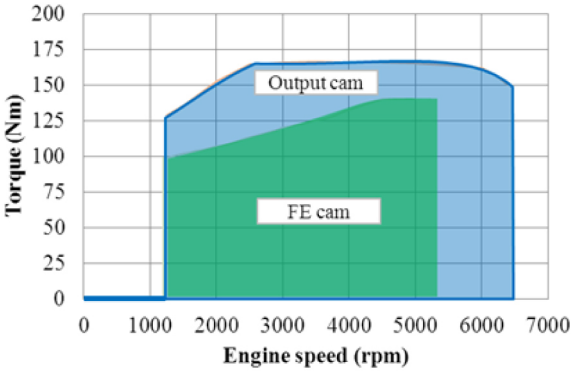


Figure 6. VTEC operation

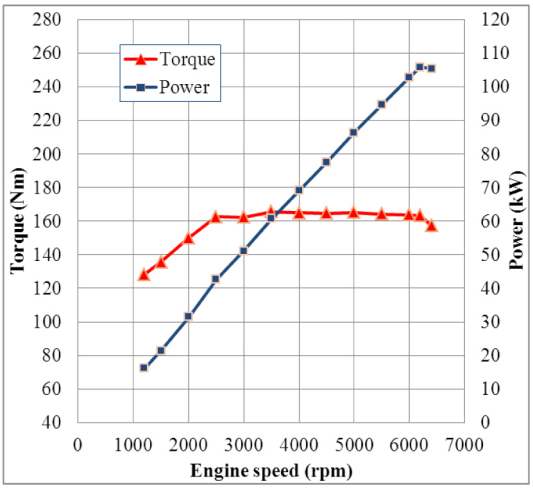


Figure 7. Output performance

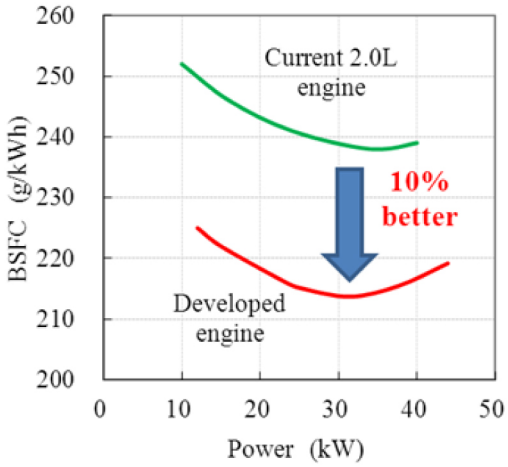


Figure 8. Enhancement of fuel consumption

e-CVT (electric coupled CVT)

A cross-sectional view of the e-CVT is shown in Figure 9. The e-CVT has a three-shaft configuration which consists of input, counter and output shaft. Transfer efficiency maximization and the miniaturization of the transmission structure are realized simultaneously by reducing the number of engagements of the gear teeth. By accelerating rotating speed of the generator compared to the engine, the generator can be operated in efficient area and its width can be reduced. As a result, the e-CVT is suitable for engine room installation since its overall width is equal or thinner than conventional transmissions.

The e-CVT has two mechanical oil pumps to deal with three driving modes of the system. One is mounted on the input shaft of the e-CVT to control the engine-drive clutch and cool the motor and generator. Another one for lubrication is mounted on the output shaft.

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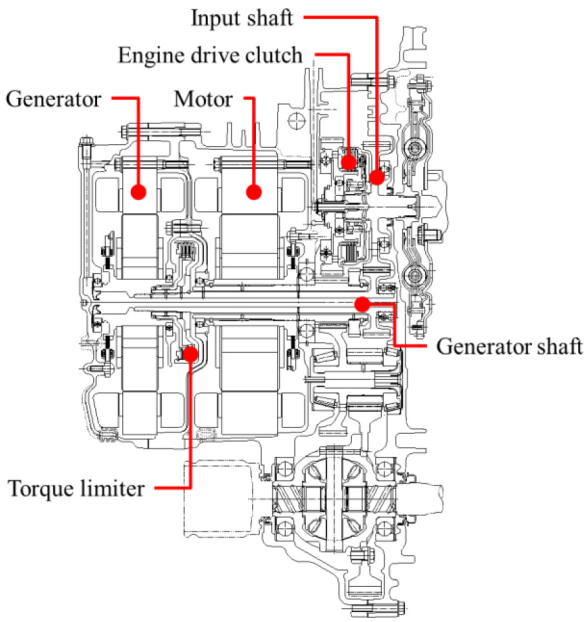


Figure 9. Cross-sectional view of e-CVT

System efficiency of "EV Drive" and regeneration is maximized since the motor output is transferred to the final gear through only two engagements of the gear teeth. Friction loss of the engine-drive clutch that switches "Hybrid Drive" and "Engine Drive" is reduced 46 to 80% compared with conventional automatic transmissions due to optimization of its clearance and the facing area.

To improve drivability of the vehicle, the engine-drive clutch is controlled in cooperation with the engine, the generator and the electric motor. If driving mode is changed from "Hybrid Drive" to "Engine drive", the speed difference between input and output is eliminated by controlling engine speed using the generator prior to the engagement of the clutch. In this way, the clutch can be engaged without noise and vibration.

Motor

A high torque and power motor with smaller size is developed by utilizing reluctance torque, high-speed rotation and voltage amplification up to 700 V to meet system feature that requires the largest driving power during "Hybrid Drive" mode. Table 2 shows achieved specifications of the motor.

Table 2. Specification of motor

Type	Synchronous motor
Max. torque [Nm]	307
Max. power [kW]	124
Max. speed [rpm]	12584

The frequency of low-load operation is high during city driving and high speed cruising. In these areas, magnet torque should be reduced and reluctance torque needs to be increased since an iron loss caused by magnetic flux fluctuation has sizable proportion.

The profile of motor core is shown in Figure 10. To maximize reluctance torque, a distributed winding motor with an IPM rotor utilizing "V" configuration magnets is developed. Figure 11 shows relationship between motor torque and current angle. Developed motor torque is enhanced 82% compared with the torque of 0 degree by utilizing reluctance torque.

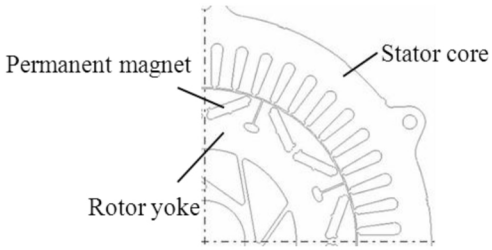


Figure 10. Profile of motor core

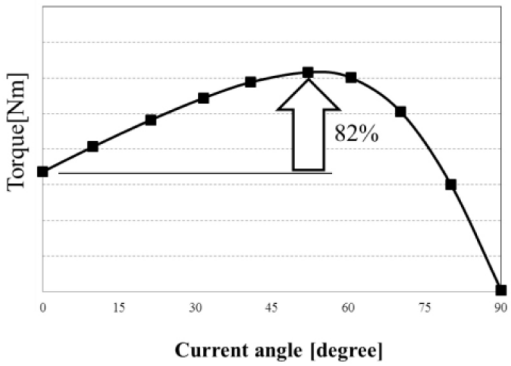


Figure 11. Torque characteristic vs. current angle

To realize high speed rotation of the motor, slit structures are made on the rotor yoke. Simulated stress distribution on the rotor with rotation is shown in Figure 12. Large stress is concentrated on the ribs outside of the magnets at the time of high speed rotation because of centrifugal forces of the permanent magnets. In order to reduce the stress on the ribs, structural rigidity around the slit is made low. As a result, stress concentration is reduced 53%.

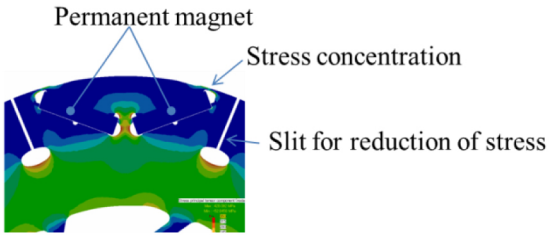


Figure 12. Simulated stress distribution

Battery voltage is amplified to realize both miniaturization and higher power of the motor. If the voltage

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of power source is fixed, the motor output power is limited at the time of high speed rotation due to increase of field weakening current and inductance. Figure 13 shows relationship between motor torque and speed at several voltage of the power source. Output power is enhanced more than 50% due to voltage amplification up to 700 V.

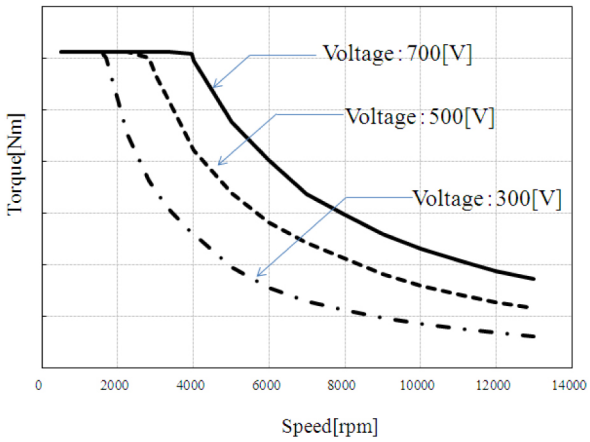


Figure 13. Torque characteristics vs. operating voltage

As a result of adopted technologies mentioned above, the motor efficiency of low-load condition is enhanced and superb fuel economy is achieved. Figure 14 shows the motor efficiency including the inverter at the best voltage.

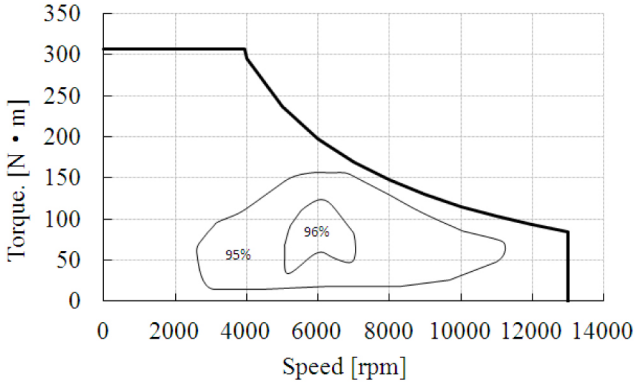


Figure 14. Motor efficiency map with the inclusion of the inverter

PCU (Power Control Unit)

Main characteristics of the PCU which controls a motor and a generator is shown in Table 3. As mentioned before, a high-torque and power motor is realized without expansion of motor size since battery voltage is amplified up to 700V.

Table 3. Specification of HEV system and PCU

Motor max output [kW]	124
Max boost voltage [V]	700
Max total output [kVA]	400
Weight [kg]	17.4
Volume [L]	11.5

Structure of the PCU is shown in Figure 15 and a block diagram is shown in Figure 16. Inverters which convert direct current into alternative current, a VCU which amplifies the battery voltage, ECUs which control a motor and a generator, various sensors and condensers constitute a PCU. Intensive arrangement of these functional parts and water cooling structure makes the overall size of the PCU small enough to install in the engine room.

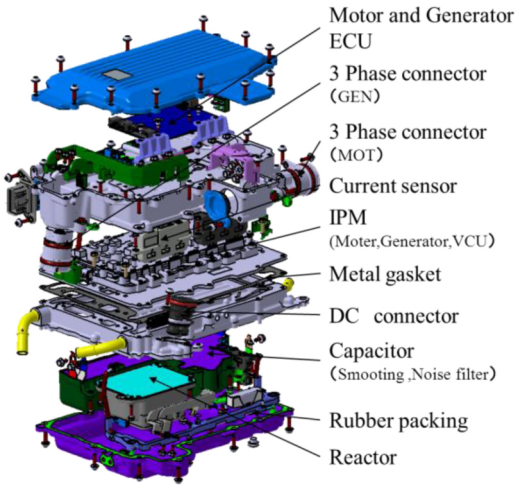


Figure 15. Structure of PCU

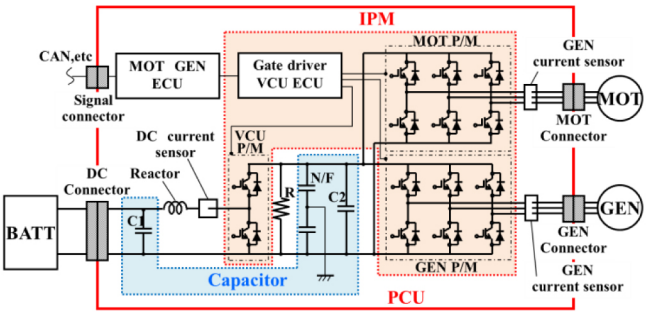


Figure 16. Block diagram of PCU

Figure 17 shows internal configuration of the IPM (Intelligent power module) which includes power modules (P/M) for the motor, generator and VCU, a gate driver, an ECU which controls a VCU and a heat sink in one package.

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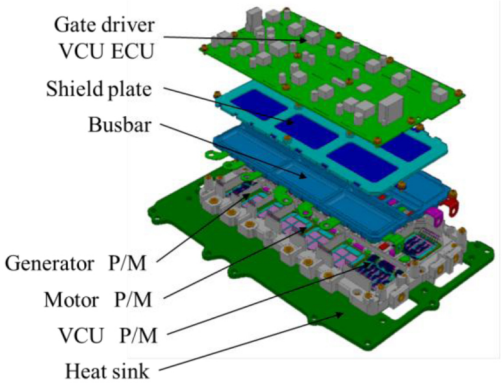


Figure 17. Configuration of IPM

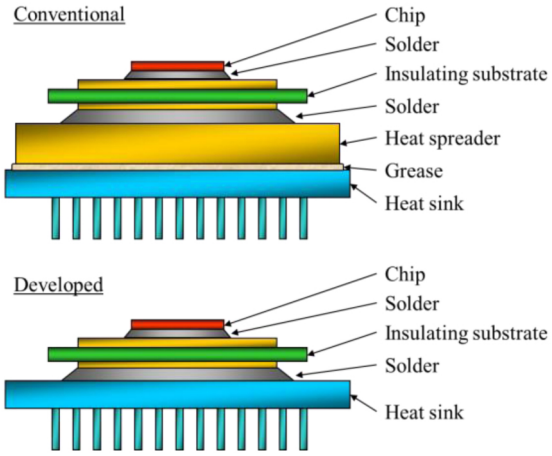


Figure 18. Structure of P/M substrate

The structure of the P/M substrate is shown in figure 18. A conventional heat spreader and grease are eliminated and insulating substrates are directly connected to the heat sink by using solder. Heat radiation is enhanced 24% due to this direct cooling structure and the number of chips can be kept to minimum as shown in Figure 19.

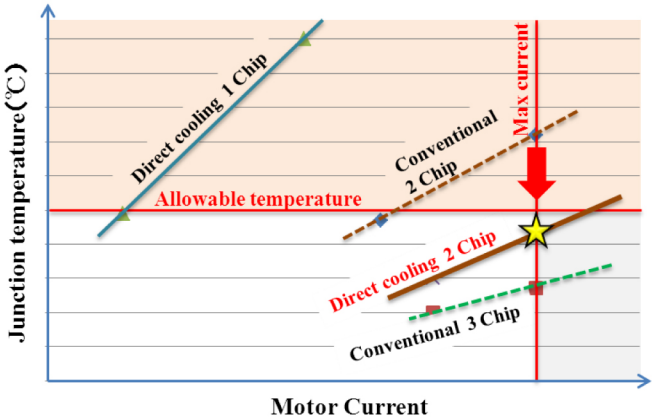


Figure 19. Comparison of P/M substrate characteristics

Li-ion battery

If the number of battery cell is increased to ensure required capacity as a plug-in HEV, system voltage is also increased. If system voltage is too high, it causes several difficulties regarding insulation. Therefore, an exclusive cell for plug-in HEV whose capacity is increased by 4.5 times compared with a conventional cell designed for usual HEV is developed. Due to 20% enhancement of the energy density, required capacity and installation suitability are satisfied simultaneously. Comparison of the battery cell specifications is shown in Table 4.

Table 4. Specification of battery cell

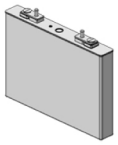
	Plug-in HYBRID	HYBRID
		
Capacity (Ah)	20.8Ah	4.6Ah
Voltage	3.2V	3.6V
Weight	0.900kg	0.276kg
SIZE (mm)	23/167/129	15.5/110/73

Figure 20 shows the configuration of the battery pack. In order to decrease the weight of the battery pack, frames and covers are made by aluminum and plates are made by magnesium. Cell voltage sensors are installed in both sides of the battery pack and monitor every cell voltage for equalization.

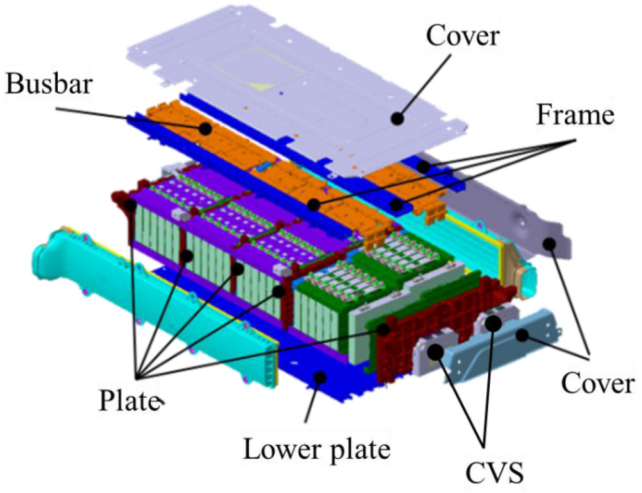


Figure 20. Battery pack

PLUG-IN CHARGING SYSTEM

An on-board charger which converts utility power into direct current for the high voltage battery charging is designed based on SAE J1772 to ensure compatibility with

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public charging stations and commercially available EVSEs (Electric Vehicle Service Equipment).

To shorten charging time and increase charged energy in a short time, output of the charger unit is determined at 6.6 kW.

In this way, electric driving range of practical usage is considered to expand. Table 5 shows main characteristics of the on-board charging system.

Table 5. Specification of On-board charging system

Charge output power		6.6 kW @ 240 VAC
Input voltage		85-264
Input frequency		45-65
Charge time	32A-240VAC	Approx. 40 min
	12A-120VAC (Genuine EVSE)	Approx. 175 min

Even though it is air-cooled, the on-board charger achieves outstanding power density by applying following technologies.

1. Reduce PFC (Power Factor Correction) output electrolytic capacitors.

2. Miniaturization of magnetic circuit by increasing switching frequency.

3. Enhancement of cooling capability of transformers and choke coils.

4. Elimination of useless space by adopting high density packaging.

The comparison of power density is shown in Figure 21.

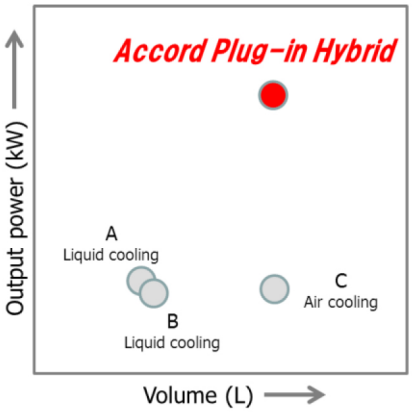


Figure 21. Power density of on-board charger

Moreover, if the vehicle is connected to EVSE, it is prohibited to operate to keep charging safety. If the charging connector is overheated due to contamination or degradation of terminals of public charging stations, a thermistor installed in the connector detects temperature increase and controls the charging current. A highly convenient and safe plug-in charging system is realized by applying these technologies.

ELECTRIC SERVO BRAKE SYSTEM

The operation of the electric servo brake system is shown in Figure 22. The brake ECU sets the target value of the brake pressure based on the pedal stroke. Thereafter, a brushless motor is driven and master cut valves are opened simultaneously to produce the brake pressure in the downstream from the slave cylinder. The brake pressure is transferred to wheel cylinders via a VSA (Vehicle Stability Assist) unit.

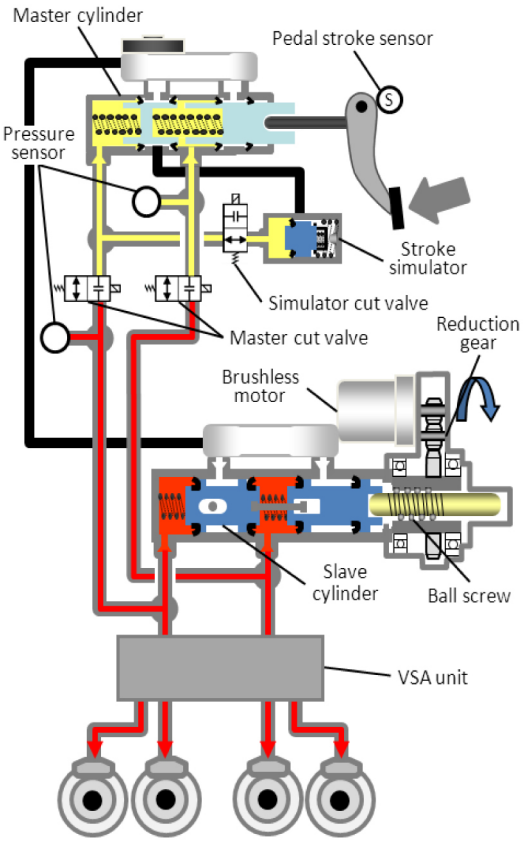


Figure 22. Block diagram of electric servo brake system

For installation in the plug-in hybrid, stall cooperative control and cooperative regeneration with the adaptive cruise control are added to the electric servo brake system developed for the BEV in order to improve fuel consumption and operability.

If the driving force and the gradient of the sloping road are in balance, the rotation of the motor stops although it is operating. If this stalled condition continues, the motor coil and drive circuits that make up the electric powertrain generate heat. However, if the driving force is reduced in order to prevent overheating, the vehicle will roll back down the sloping road. Stall cooperative control prevents both overheating and rolling back by generating friction braking force using the electric servo brake system.

Cooperative regeneration with the adaptive cruise control distributes regeneration brake and friction brake properly when acceleration and deceleration is controlled

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automatically to keep a distance between vehicles by using radar sensor.

SYSTEM CONTROL

Plug-in hybrid operation

Figure 23 shows an outline of plug-in hybrid operation.

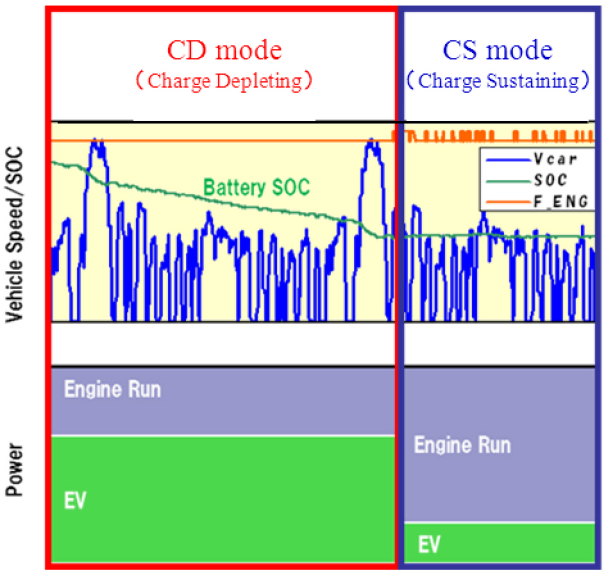


Figure 23. Plug-in hybrid operation

If the battery is fully charged via plug-in charge, CD (Charge Depleting) mode is selected and the vehicle is propelled by utilizing stored electric energy. Since EV capability is determined by the performance of the on-board high voltage battery, the battery specification is fixed by considering following targets.

- Whole of city driving should be covered by "EV Drive".
- EV range should be over 10 miles.

If the battery SOC (State of Charge) drops to predetermined level, CS (Charge Sustaining) mode is automatically selected as shown in Figure 23. An outline of CS mode which is a basic operation of this system is shown in Figure 24.

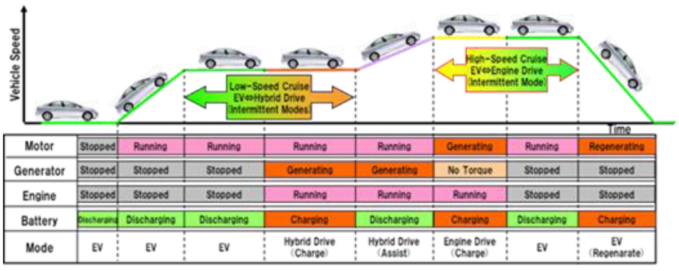


Figure 24. CS (Charge Sustaining) operation

During low-load situation such as launching or city driving, "EV Drive" is mainly selected. Then, driving mode is switched to "Hybrid Drive" for the acceleration during normal-load or heavy-load situation. If the vehicle is cruising, intermittent operation between "EV Drive" and "Hybrid Drive" or "Engine drive" is selected. The high voltage battery is charged during engine activation such as "Hybrid Drive" and "Engine Drive". Then, stored electric energy is used in "EV Drive" mode afterwards. Since thermal efficiency of the engine is predominant to overall system efficiency, the key to improve overall fuel economy is highly efficient operation of the engine. Figure 25 shows operating area of three driving modes.

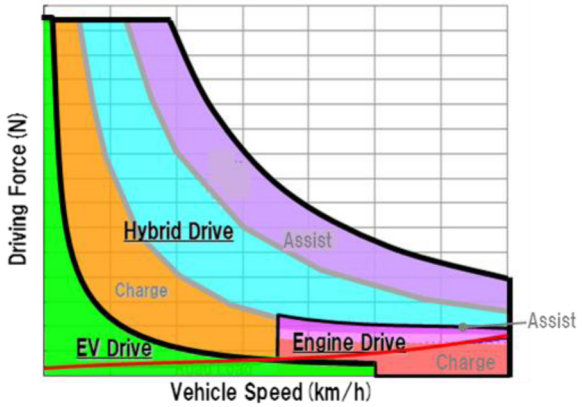


Figure 25. Operating area of three driving modes (Charge sustaining)

This hybrid system can provide whole of required propulsion power of all driving situation by the electric motor. Therefore, three driving modes can be selected depending on the driving load. If the system efficiency can be enhanced by transferring engine output directly to the wheels like high speed cruising, the engine drive clutch is engaged in order to select "Engine Drive" mode.

A major point of difference between CS and CD mode is explained in below.

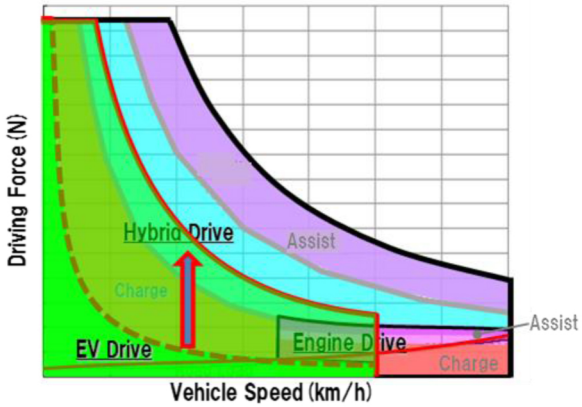


Figure 26. Operating area of three driving modes (Charge depleting)

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Operating area of "EV Drive" in CD mode is shown in Figure 26. "EV Drive" area is enlarged to cover whole city driving. Except enlargement of "EV Drive", "Hybrid Drive" and "Engine Drive" are switched just like in CS mode.

Engine operation

Figure 27 shows an outline of the engine operating points. The engine is activated in "Hybrid Drive" mode and "Engine Drive" mode with this system. An outline of the engine operation in these two modes is explained in a following section.

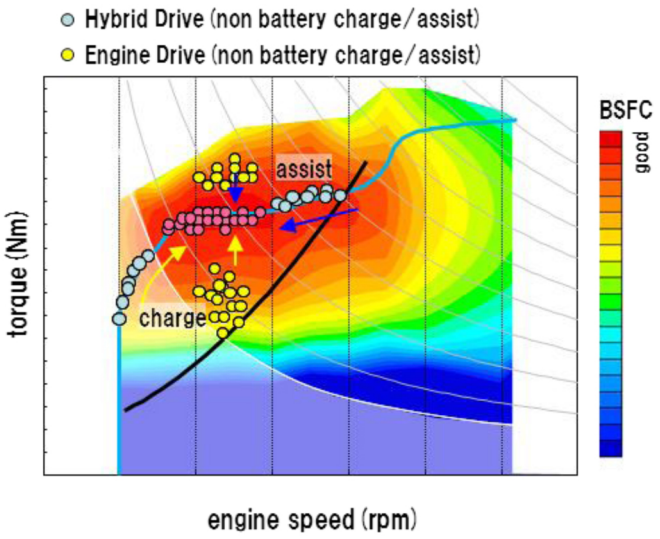


Figure 27. Engine operating point

In "Hybrid Drive" mode, there is no restriction regarding rotational speed of the wheels and the engine, since they have no physical connection each other. Therefore, engine speed can be determined arbitrarily without any relationship with vehicle speed. If the relationship between input (engine speed) and output (motor speed) shaft of the e-CVT is considered, it can be operated just like a conventional CVT.

The engine operating point (speed and torque) is determined to follow the minimum BSFC for every possible power demand. Moreover, the engine load is increased or decreased by charging or discharging the high voltage battery in order to move engine operating point toward the higher efficiency point on the minimum BSFC line.

In "Engine Drive" mode, engine speed is uniquely fixed if vehicle speed is determined since the engine is physically connected to the wheels via fixed reduction gear. The relationship between engine speed and torque during cruise on flat road is shown as a black line in Figure 27. Although "Engine Drive" mode is mainly utilized during high speed cruise, engine operating points are separated from the minimum BSFC line toward lower torque area at the low vehicle speed side. Therefore, engine load adjustment by charging or discharging the high voltage battery is also performed in this mode to fix engine operating points in highly efficient area.

Intermittent operation

Fuel economy enhancement effect of the intermittent operation between "EV Drive" and "Hybrid Drive" or "Engine Drive" is described in this section.

Figure 28 shows fuel economy enhancement of intermittent operation between "EV Drive" and "Hybrid Drive". If required driving power is lower than 15 kW, fuel economy is enhanced up to 50% compared with continuous "Hybrid Drive" mode (red line in figure 28) by applying intermittent operation which means the high voltage battery is charged during "Hybrid Drive" mode and stored electric energy is used in "EV Drive" mode afterwards. On the other hand, fuel economy enhancement effect is reduced or gets worse, if required driving power is over 15 kW. Permitted area of intermittent operation is determined by the contribution to fuel economy enhancement.

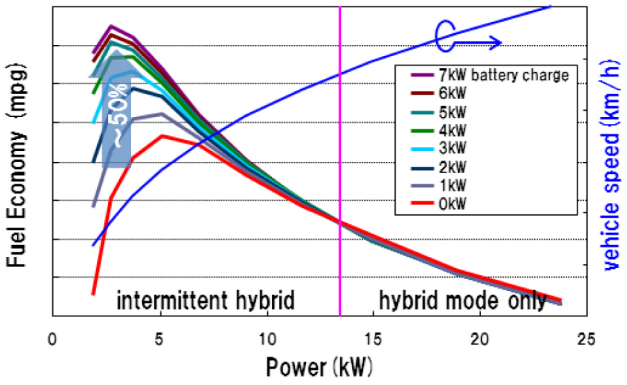


Figure 28. Fuel economy enhancement of intermittent operation

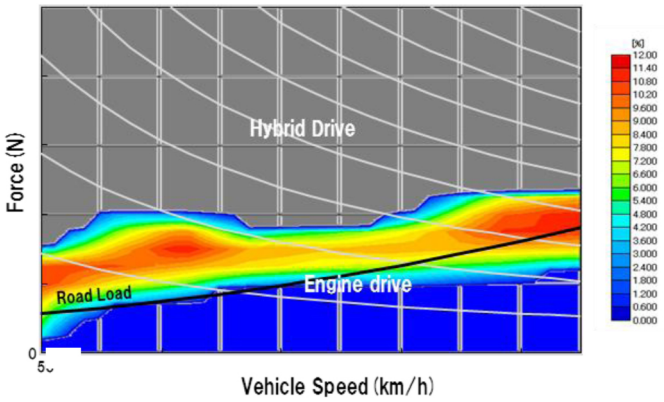


Figure 29. Comparison of fuel consumption of "Hybrid Drive" and "Engine Drive"

Comparison of fuel consumption of "Hybrid Drive" and "Engine Drive" is shown in Figure 29. A black line in Figure 29 represents the road load during driving on a flat road. "Engine drive" has superior fuel economy compared with "Hybrid Drive" in red area, and both driving modes have similar efficiency in blue area. This diagram represents that

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the efficiency of "Engine Drive" is better than "Hybrid Drive" in the area of cruise and slight acceleration. Based on this relationship, "Hybrid Drive" and "Engine drive" are switched.

Energy management

In order for the vehicle to operate according to the mode of operation discussed in the previous sections, the energy management of the high voltage battery is important. Over-discharging and overcharging of the high voltage battery are big technical issues in all driving condition because they not only cause lower fuel economy but decrease the battery durability.

In each driving mode, discharging and charging of the high voltage battery is controlled properly from fuel economy, drivability and durability point of view based on the state of charge (SOC). Battery SOC is managed in appropriate area while maximizing system efficiency as mentioned above.

Output and input limitations of the high voltage battery in each operation mode are shown in Figure 30. This power management setting allows choosing "EV Drive" in most cases of CD mode. To avoid fuel efficiency deterioration due to wasteful charging, charging in "Hybrid Drive" and "Engine Drive" is permitted in CS mode only. To recover stored electricity amount, electric motor assist is inhibited and the high voltage battery is charged forcibly in the low SOC situation. Also, the battery is charged as much as possible except full charge condition during regeneration.

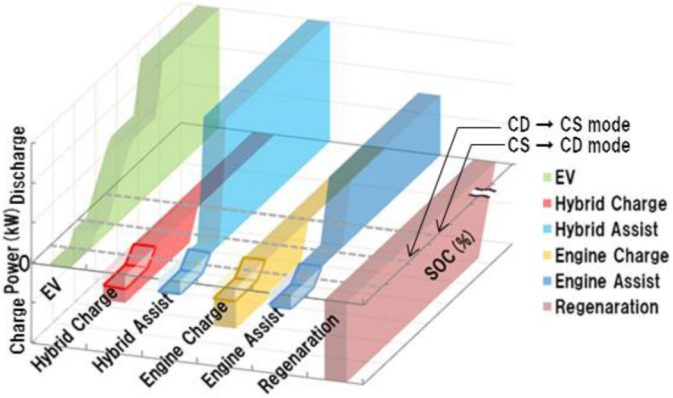


Figure 30. Battery power management

In actual energy management, permitted output is determined in real time within the confine of charge and discharge limit as shown in Figure 30. Vehicle speed, engine rpm and SOC transition in actual driving cycle with proposed energy management concept are shown in Figure 31. Vehicle speed is indicated in 3 different colors, "EV Drive (Green)", "Hybrid Drive (Red)" and "Engine Drive (Gray)".

"Hybrid Drive" is selected if strong acceleration is needed and "EV Drive" is selected during slight acceleration or deceleration. As mentioned before, intermittent operation between "Engine Drive" and "EV Drive" is performed during slight acceleration or cruise in a high speed situation. Energy management control is optimized by choosing proper operating mode in response to the driving condition and adopting engine load adjustment during engine activation.

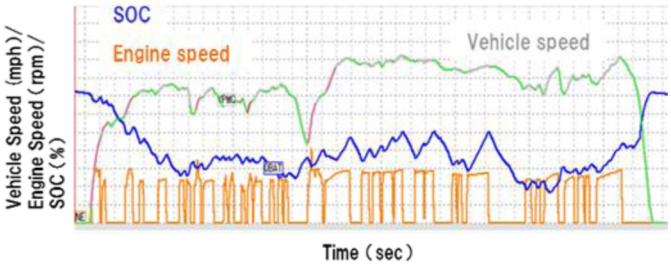


Figure 31. Driving data

SULEV20

As in the case of fuel economy enhancement, to meet a strict emission standard is an important issue for environmental protection. Figure 32 shows the exhaust system of this vehicle.

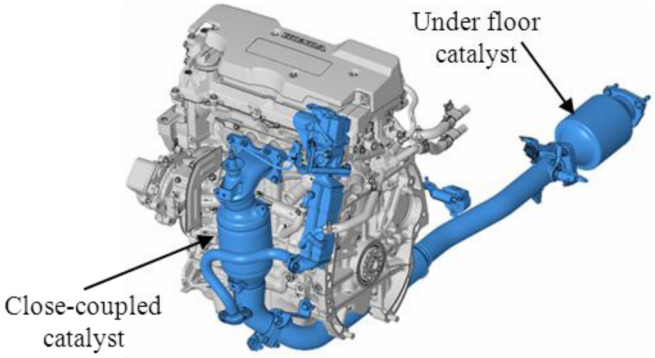


Figure 32. Exhaust system

The new catalyst warm-up control is realized by using electric powertrain cleverly. Right after the engine start, generation of electricity is kept at lower level to reduce feed emission through the engine load reduction. After the close-coupled catalyst warm-up, the engine load is increased by generating electricity to warm up the under-floor catalyst.

As a result of these new control methods, this vehicle meets SULEV20 standard.

RESULT OF FUEL ECONOMY ENHANCEMENT

Contributions to fuel economy enhancement of each technical item of the proposed two-motor hybrid system compared to an imaginary conventional gasoline-powered vehicle equipped with a four-stroke gasoline engine without special features such as Atkinson cycle or cooled EGR are shown in Figure 33.

Fuel economy is enhanced approximately 39% by hybridization utilizing developed two-motor hybrid system.

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BSFC of the engine is improved approximately 10% by adopting cooled EGR system and VTEC technology to realize Atkinson cycle. The efficiency enhancement of the motor and generator including PCU contributes approximately 10% of overall efficiency by the effective use of the reluctance torque and the optimization of the motor operating voltage using VCU. Brake loss reduction as a result of the electric servo brake system application realizes 8% overall efficiency enhancement compared with a conventional regeneration brake system. Moreover, low rolling resistance tires and aerodynamic drag decrease contributes approximately 8% of overall efficiency enhancement.

As a result, fuel economy is enhanced approximately 70% compared with a conventional vehicle.

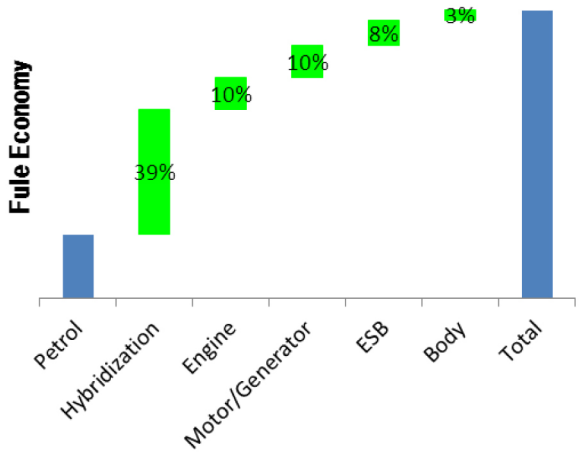


Figure 33. Contributions to fuel economy enhancement of the proposed two-motor hybrid system

SUMMARY/CONCLUSIONS

The following fuel consumption values and compliances with emission regulations are achieved while maintaining sufficient driving performance and commodiousness as a mid-class sedan by installing a newly developed two-motor plug-in hybrid system which consists of an electric coupled CVT, an integrated PCU, a high capacity Li-ion battery and a high-power on-board charger into 2014 Accord plug-in hybrid combined with an Atkinson cycle engine and an electric servo brake system.

1. EV range: 13 mile

2. Fuel economy:

• Charge depleting: 115 MPGe

• Charge sustaining: 46 MPG

3. Compliance with LEV3-SULEV20

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2. Ohkubo, N., Matsushita, S., Ueno, M., Akamine, K. et al., "Application of Electric Servo Brake System to Pulg-In Hybrid Vehicle," *SAE Int. J. Passeng. Cars – Electron. Electr. Syst.* 6(1):255-260, 2013, doi: 10.4271/2013-01-0697.

DEFINITIONS/ABBREVIATIONS

AER - All Electric Range

BEV - Battery Electric Vehicle

CD mode - Charge Depleting mode

CS mode - Charge Sustaining mode

CVT - Continuously Variable Transmission

DOHC - Double Over Head Camshaft

ECU - Electric Control Unit

e-CVT - electric coupled CVT

EGR - Exhaust Gas Recirculation system

EV - Electric Vehicle

EVSE - Electric Vehicle Service Equipment

FCEV - Fuel Cell Electric Vehicle

HEV - Hybrid Electric Vehicle

IPM - Intelligent Power Module

IPU - Intelligent Power Unit

MPGe - Miles Per Gallon equivalent

PCU - Power Control Unit

PFC - Power Factor Correction

P/M - Power Module

VCU - Voltage Control Unit

VSA - Vehicle Stability Assist

VTEC - Variable valve Timing and Electric Control